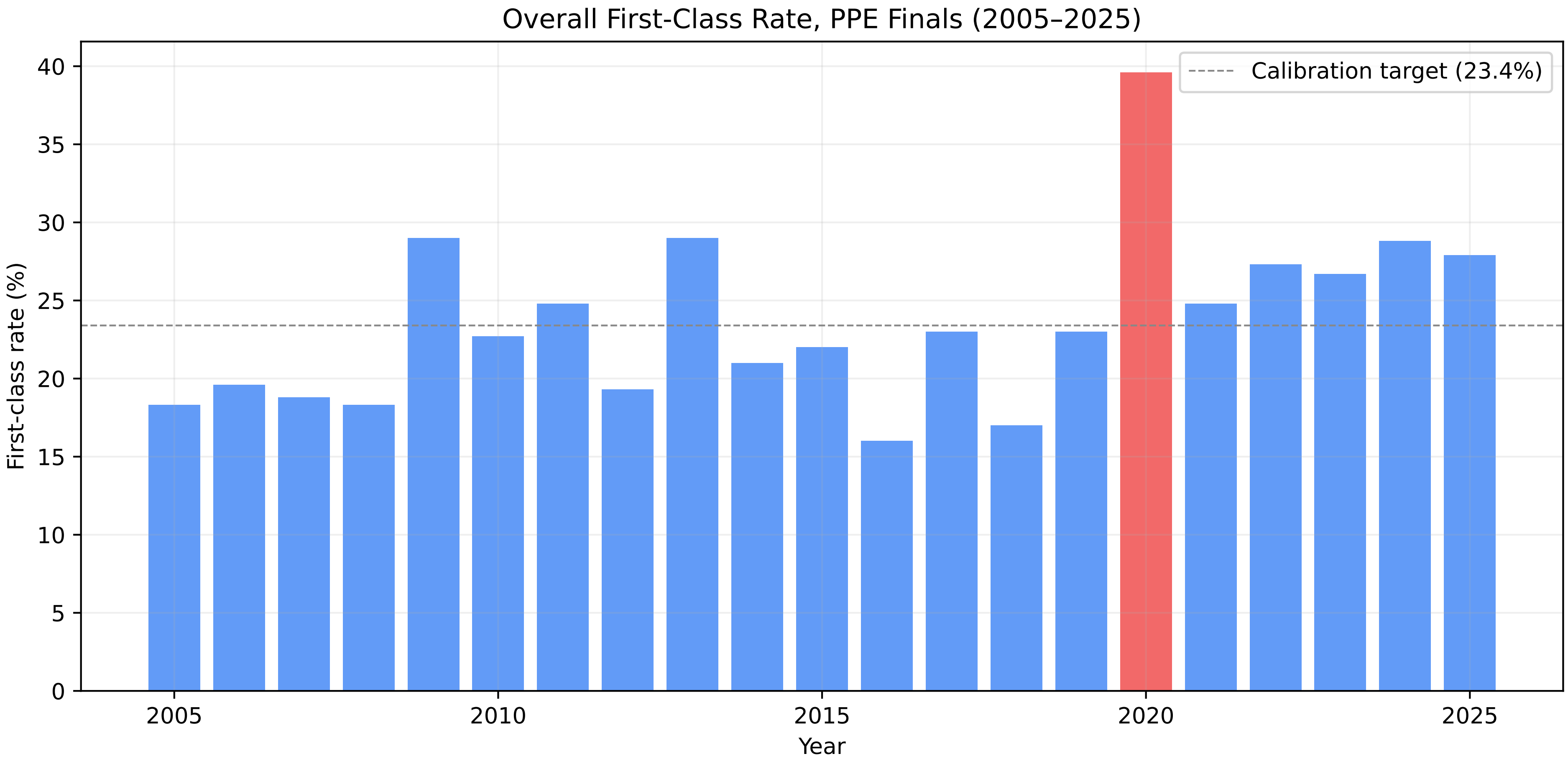
Oxford PPE Finals

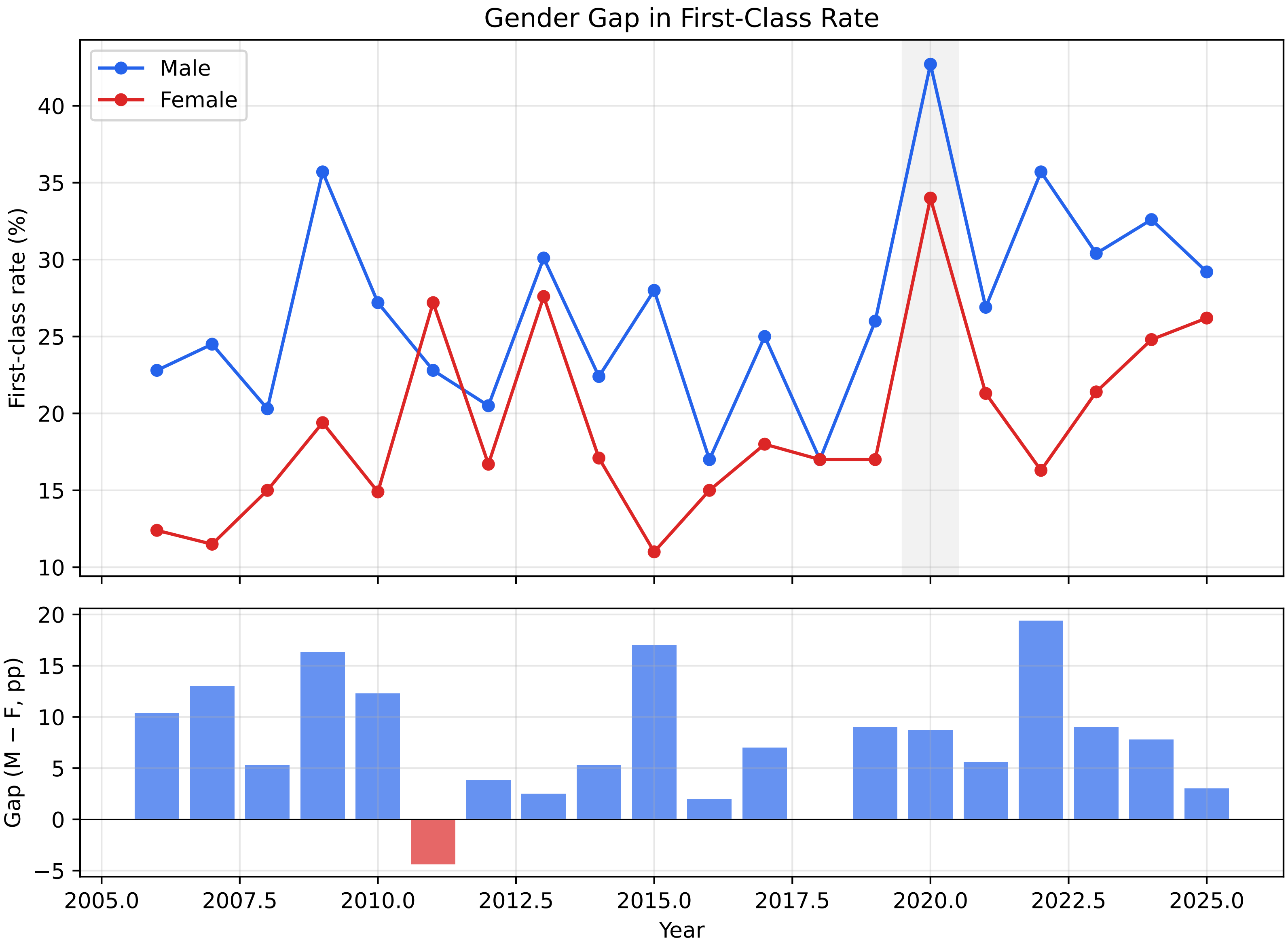
Statistical Analysis of Examiners' Reports 2011–2025

81 papers fitted · 100k Monte Carlo simulations · 15 years of data

$\sigma_{ability}$ = 2.58 | Calibration target: 23.4% first rate

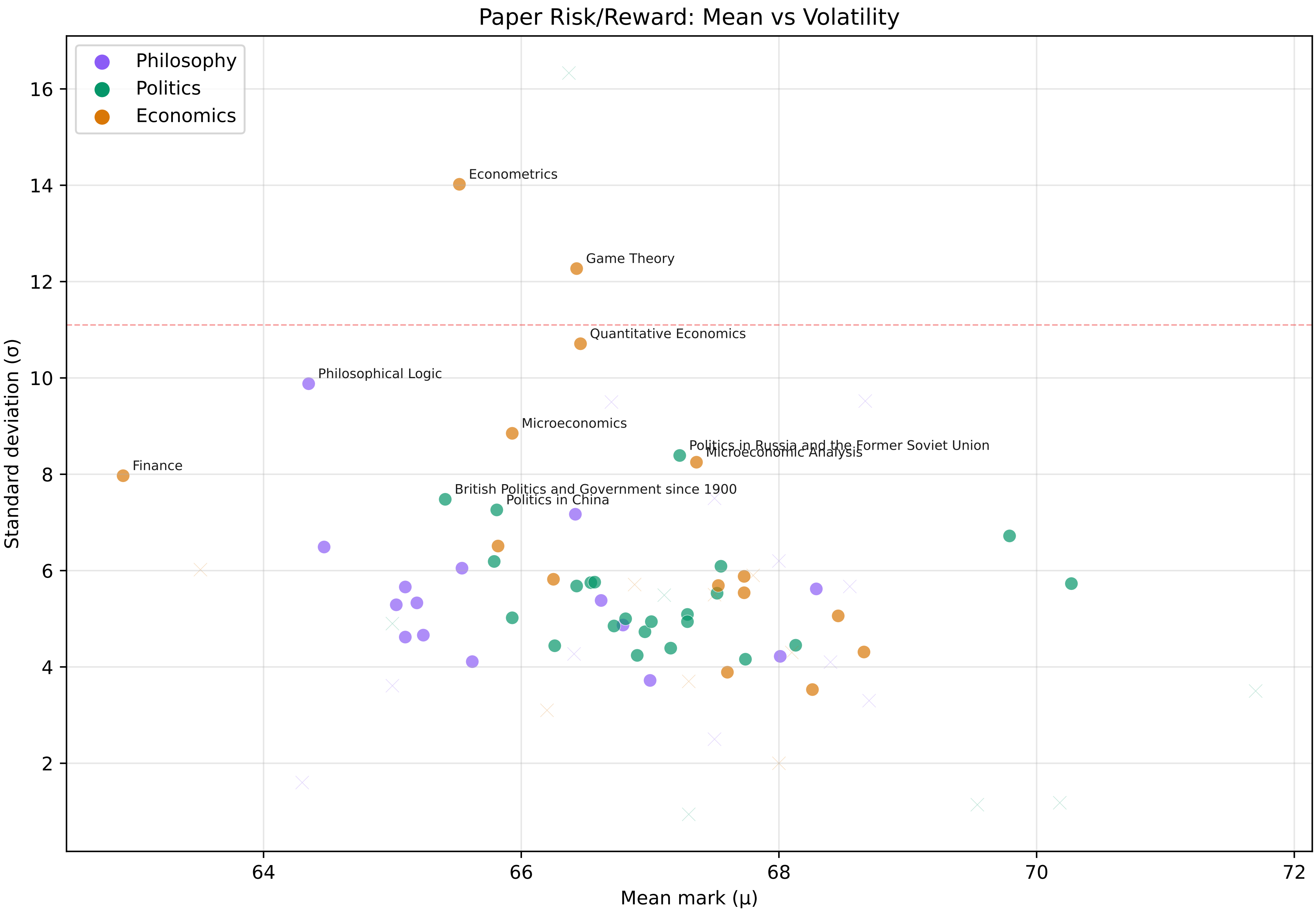


Overall First-Class Rate, PPE Finals (2005–2025)
Calibration target (23.4%)
First-class rate (%)
0
5
10
15
20
25
30
35
40
2005
2010
2015
2020
2025
Year


Gender Gap in First-Class Rate
Male
Female
First-class rate (%)
10
15
20
25
30
35
40
Gap (M − F, pp)
20
15
10
5
0
−5
2005.0
2007.5
2010.0
2012.5
2015.0
2017.5
2020.0
2022.5
2025.0
Year

Subject-Level Summary

Subject	Mean	SD	Papers	Total n	First Rate	Within Var	Between Var	Ratio
Philosophy	65.5	5.51	26	2831	20.7%	31.51	0.94	33.5
Politics	66.76	5.56	30	5020	28.1%	32.43	0.94	34.4
Economics	66.33	8.22	23	4210	31.9%	73.88	0.82	89.9



Paper Risk/Reward: Mean vs Volatility
Philosophy
Politics
Economics
Econometrics
Game Theory
Quantitative Economics
Philosophical Logic
Microeconomics
Politics in Russia and the Former Soviet Union
Microeconomic Analysis
Finance
British Politics and Government since 1900
Politics in China
Standard deviation (σ)
Mean mark (μ)
16
14
12
10
8
6
4
2
64
66
68
70
72

Highest Mean (Easiest Papers)

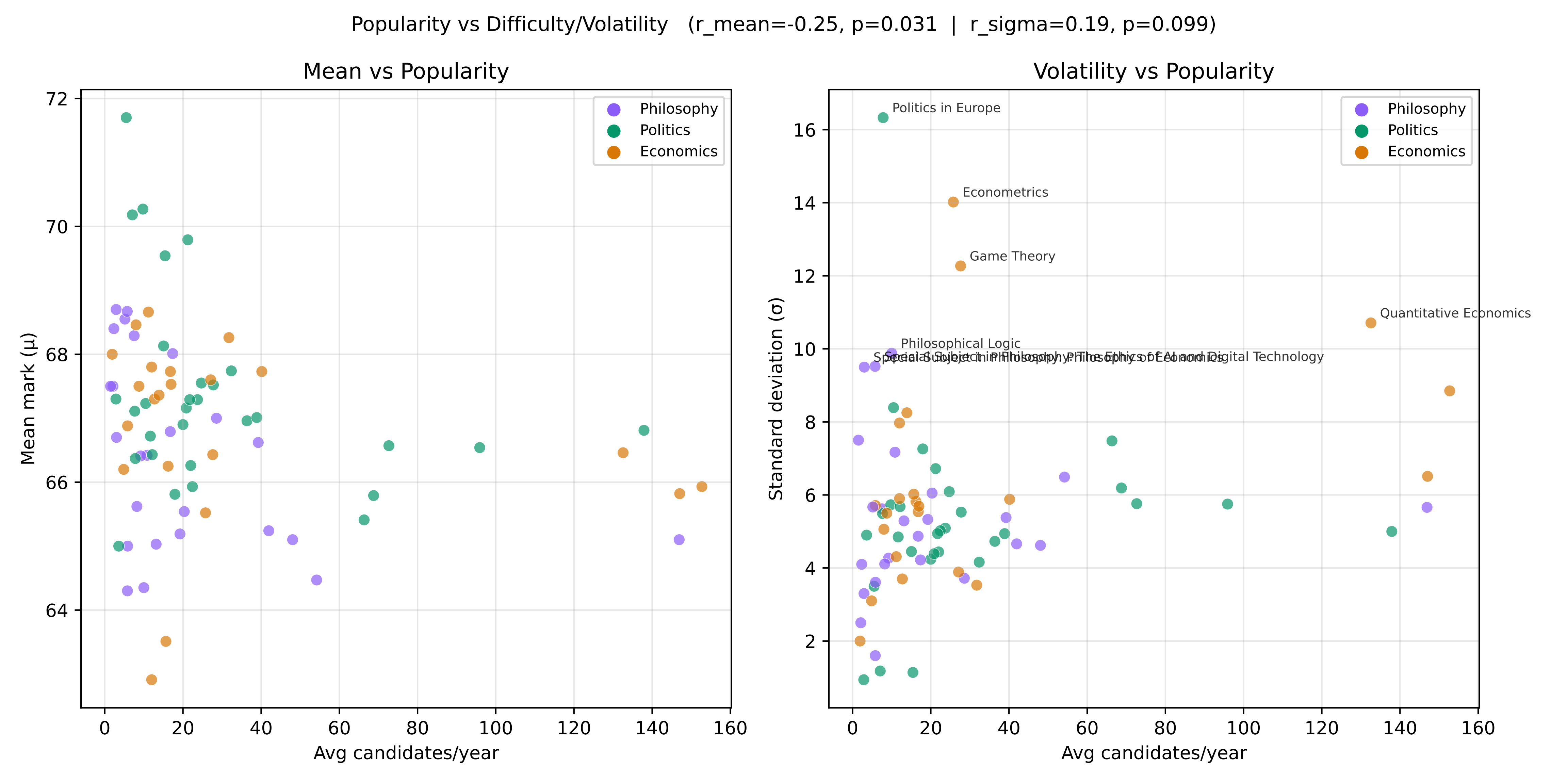
#	Paper	Subject	μ	σ	%1st	%<50
1	Supervised Dissertation in Politics	Politics	71.7	3.5	68.6	0.0
2	Politics in South Asia	Politics	70.3	5.7	51.9	0.0
3	The Politics of the European Union	Politics	70.2	1.2	56.1	0.0
4	Thesis in Politics	Politics	69.8	6.7	48.8	0.2
5	International Relations in the Era of Two Wor	Politics	69.5	1.1	34.3	0.0
6	Philosophy of Cognitive Science	Philosophy	68.7	3.3	34.7	0.0
7	Special Subject in Philosophy: The Ethics of	Philosophy	68.7	9.5	44.4	2.5
8	Special Subject in Economics: Environmental E	Economics	68.7	4.3	37.8	0.0
9	The Philosophy of Wittgenstein	Philosophy	68.5	5.7	39.9	0.1
10	Behavioural and Experimental Economics	Economics	68.5	5.1	38.0	0.0
11	Philosophy of Science	Philosophy	68.4	4.1	34.8	0.0
12	Thesis in Philosophy	Philosophy	68.3	5.6	38.0	0.1
13	Economics of Developing Countries	Economics	68.3	3.5	31.1	0.0
14	Marx and Marxism	Politics	68.1	4.5	33.7	0.0
15	The Philosophy and Economics of the Environme	Economics	68.1	4.3	32.9	0.0

Lowest Mean (Hardest Papers)

#	Paper	Subject	μ	σ	%1st	%<50
1	Finance	Economics	62.9	8.0	18.7	5.3
2	Labour Economics and Industrial Relations	Economics	63.5	6.0	14.1	1.2
3	Philosophy of Science and Social Science	Philosophy	64.3	1.6	0.0	0.0
4	Philosophical Logic	Philosophy	64.3	9.9	28.4	7.3
5	Knowledge and Reality	Philosophy	64.5	6.5	19.7	1.3
6	Quantitative Methods in Politics and Sociolog	Politics	65.0	4.9	15.4	0.1
7	The Philosophy of Kant	Philosophy	65.0	3.6	8.3	0.0
8	Philosophy of Mind	Philosophy	65.0	5.3	17.4	0.2
9	Ethics	Philosophy	65.1	5.7	19.3	0.4
10	Early Modern Philosophy	Philosophy	65.1	4.6	14.4	0.1
11	Aesthetics and the Philosophy of Criticism	Philosophy	65.2	5.3	18.3	0.2
12	Plato: Republic	Philosophy	65.2	4.7	15.4	0.1
13	British Politics and Government since 1900	Politics	65.4	7.5	27.0	2.0
14	Econometrics	Economics	65.5	14.0	37.0	13.5
15	Aristotle: Nicomachean Ethics	Philosophy	65.5	6.0	23.1	0.5

Highest Volatility (σ)

#	Paper	Subject	μ	σ	%1st	%<50
1	Politics in Europe	Politics	66.4	16.3	40.0	16.1
2	Econometrics	Economics	65.5	14.0	37.0	13.5
3	Game Theory	Economics	66.4	12.3	38.4	9.1
4	Quantitative Economics	Economics	66.5	10.7	37.0	6.2
5	Philosophical Logic	Philosophy	64.3	9.9	28.4	7.3
6	Special Subject in Philosophy: The Ethics of	Philosophy	68.7	9.5	44.4	2.5
7	Special Subject in Philosophy: Philosophy of	Philosophy	66.7	9.5	36.4	3.9
8	Microeconomics	Economics	65.9	8.8	32.3	3.6
9	Politics in Russia and the Former Soviet Unio	Politics	67.2	8.4	37.1	2.0
10	Microeconomic Analysis	Economics	67.4	8.2	37.4	1.8
11	Finance	Economics	62.9	8.0	18.7	5.3
12	Frege, Russell, and Wittgenstein	Philosophy	67.5	7.5	36.9	1.0
13	British Politics and Government since 1900	Politics	65.4	7.5	27.0	2.0
14	Politics in China	Politics	65.8	7.3	28.2	1.5
15	Philosophy of Logic and Language	Philosophy	66.4	7.2	30.9	1.1

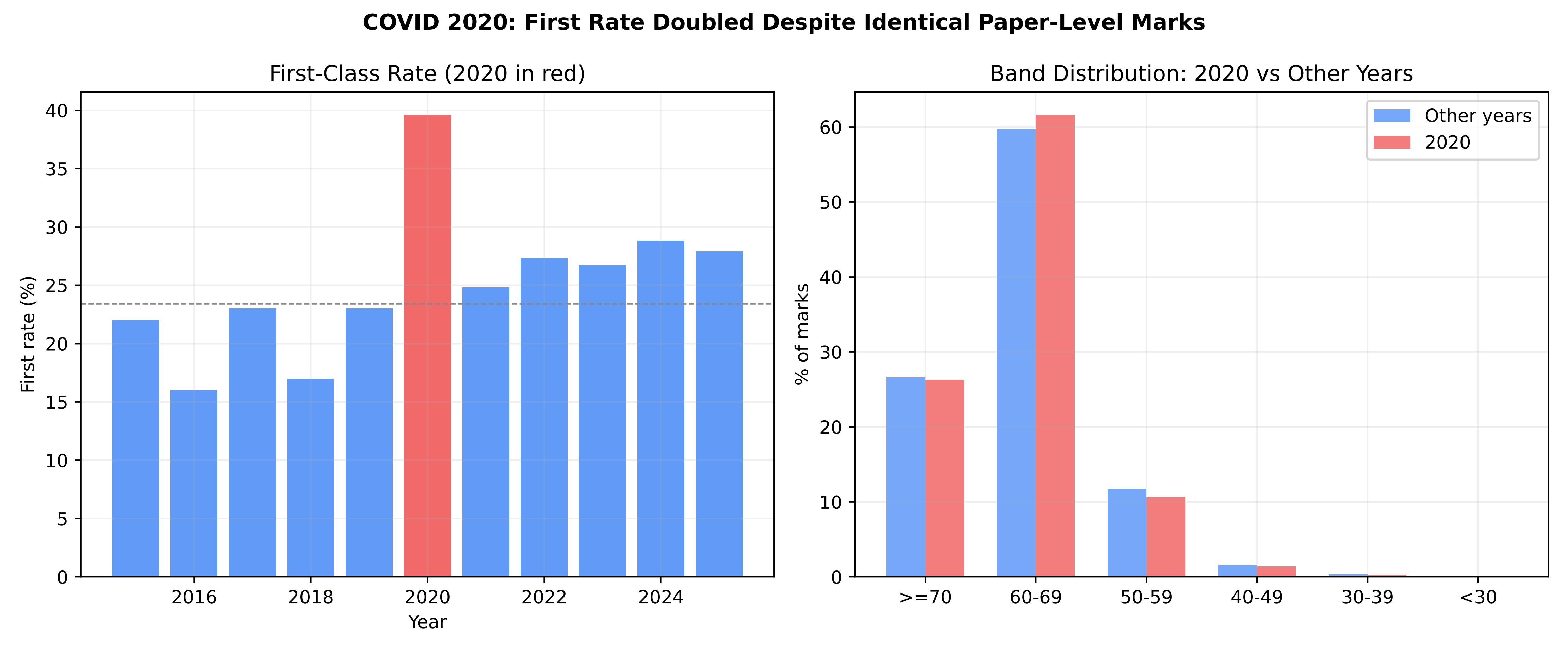

Popularity vs Difficulty/Volatility (r_mean=-0.25, p=0.031 | r_sigma=0.19, p=0.099)
Mean vs Popularity
Mean mark (μ)
Avg candidates/year
Philosophy
Politics
Economics
Volatility vs Popularity
Standard deviation (σ)
Avg candidates/year
Politics in Europe
Econometrics
Game Theory
Quantitative Economics
Philosophical Logic

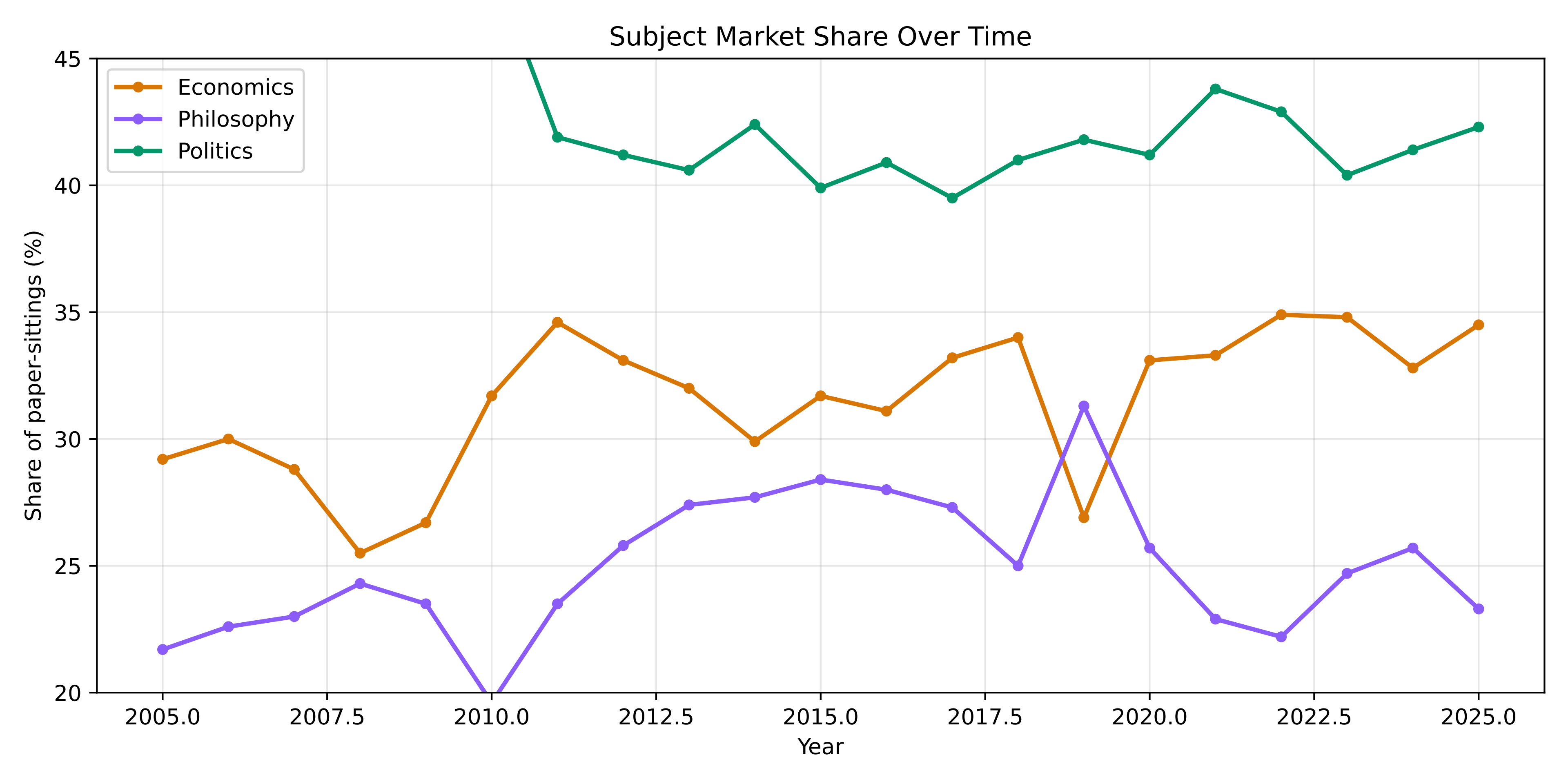
Temporal Trends in Paper Mean Marks ($p < 0.10$)

Paper	Subject	Slope	95% CI	p-value	R^2	n
Quantitative Economics	Economics	+0.549	[+0.300, +0.798]	0.0012	0.795	9
Microeconomic Analysis	Economics	+1.651	[+0.817, +2.485]	0.0053	0.883	6
Political Thought: Plato to Rousseau	Politics	+0.235	[+0.092, +0.379]	0.0061	0.681	9
Political Thought: Bentham to Weber	Politics	+0.353	[+0.114, +0.592]	0.0100	0.636	9
Philosophical Logic	Philosophy	-0.629	[-1.094, -0.163]	0.0152	0.593	9
Politics in South Asia	Politics	+0.473	[+0.118, +0.828]	0.0162	0.586	9
Public Economics	Economics	+0.255	[+0.022, +0.487]	0.0358	0.490	9
Game Theory	Economics	+0.419	[-0.010, +0.849]	0.0541	0.433	9
Politics in China	Politics	-0.405	[-0.841, +0.030]	0.0637	0.409	9
Money and Banking	Economics	-0.194	[-0.406, +0.017]	0.0664	0.403	9
Social Policy	Politics	+0.236	[-0.030, +0.502]	0.0743	0.386	9
Comparative Government	Politics	+0.114	[-0.026, +0.254]	0.0948	0.347	9
Government and Politics of the United St	Politics	+0.331	[-0.079, +0.741]	0.0976	0.343	9

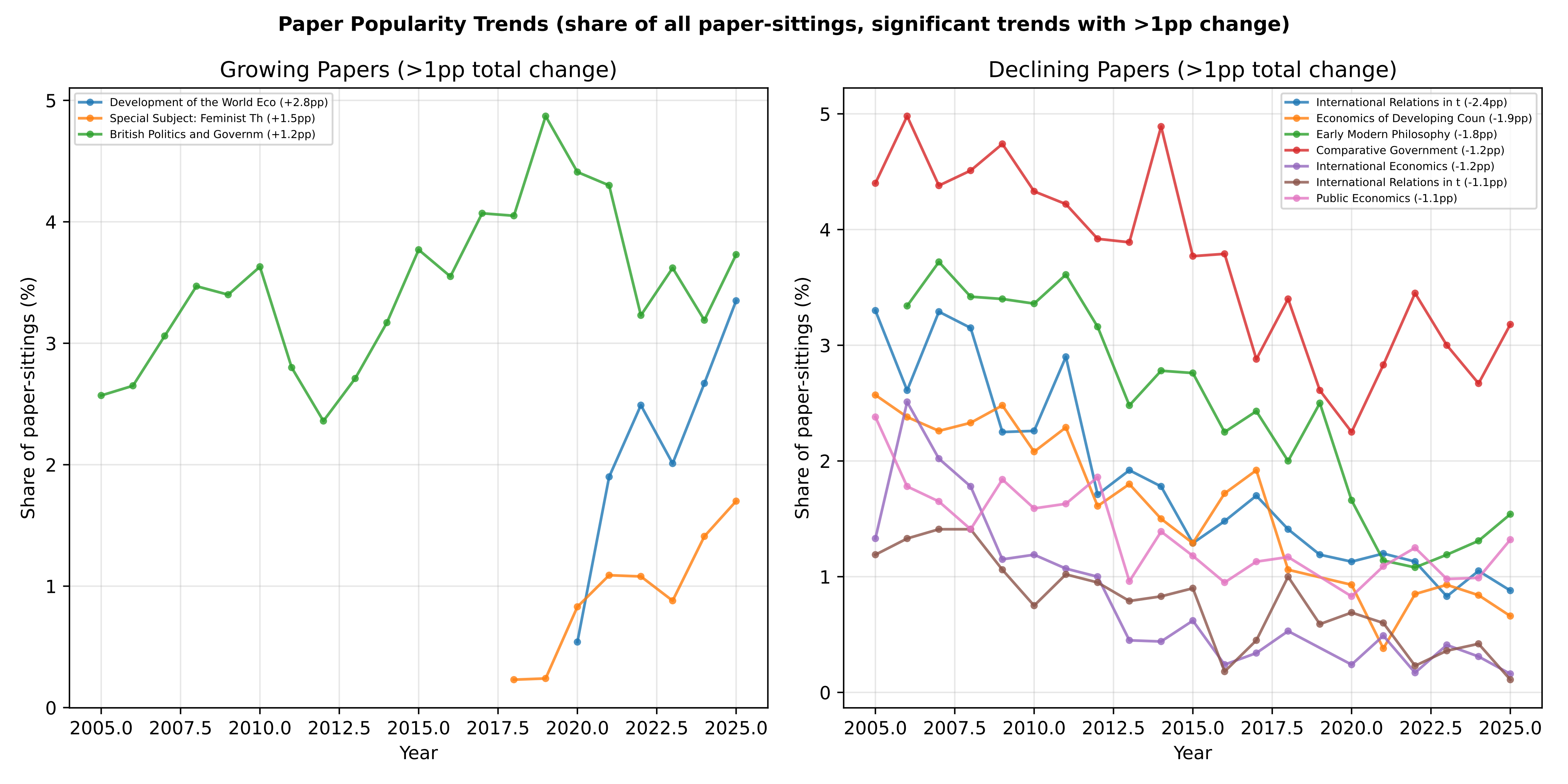
Marginal Paper Value: Effect of 8th Paper on P(1st)
Baseline = 23.3% with British Politics and Government since 1900

Paper	Subject	μ	σ	P(1st)	ΔP(1st)
Supervised Dissertation in Politics	Politics	71.7	3.5	30.0%	+6.8%
Politics in South Asia	Politics	70.3	5.7	28.5%	+5.2%
The Politics of the European Union	Politics	70.2	1.2	28.1%	+4.8%
Thesis in Politics	Politics	69.8	6.7	28.0%	+4.8%
International Relations in the Era of Tw	Politics	69.5	1.1	27.3%	+4.0%
Special Subject in Philosophy: The Ethic	Philosophy	68.7	9.5	27.2%	+3.9%
The Philosophy of Wittgenstein	Philosophy	68.5	5.7	26.5%	+3.2%
Special Subject in Economics: Environmen	Economics	68.7	4.3	26.4%	+3.2%
Philosophy of Cognitive Science	Philosophy	68.7	3.3	26.3%	+3.1%
Behavioural and Experimental Economics	Economics	68.5	5.1	26.3%	+3.0%
...					
The Philosophy of Kant	Philosophy	65.0	3.6	22.1%	-1.1%
Knowledge and Reality	Philosophy	64.5	6.5	22.0%	-1.2%
Philosophy of Science and Social Science	Philosophy	64.3	1.6	21.4%	-1.9%
Labour Economics and Industrial Relation	Economics	63.5	6.0	21.0%	-2.3%
Finance	Economics	62.9	8.0	20.6%	-2.6%

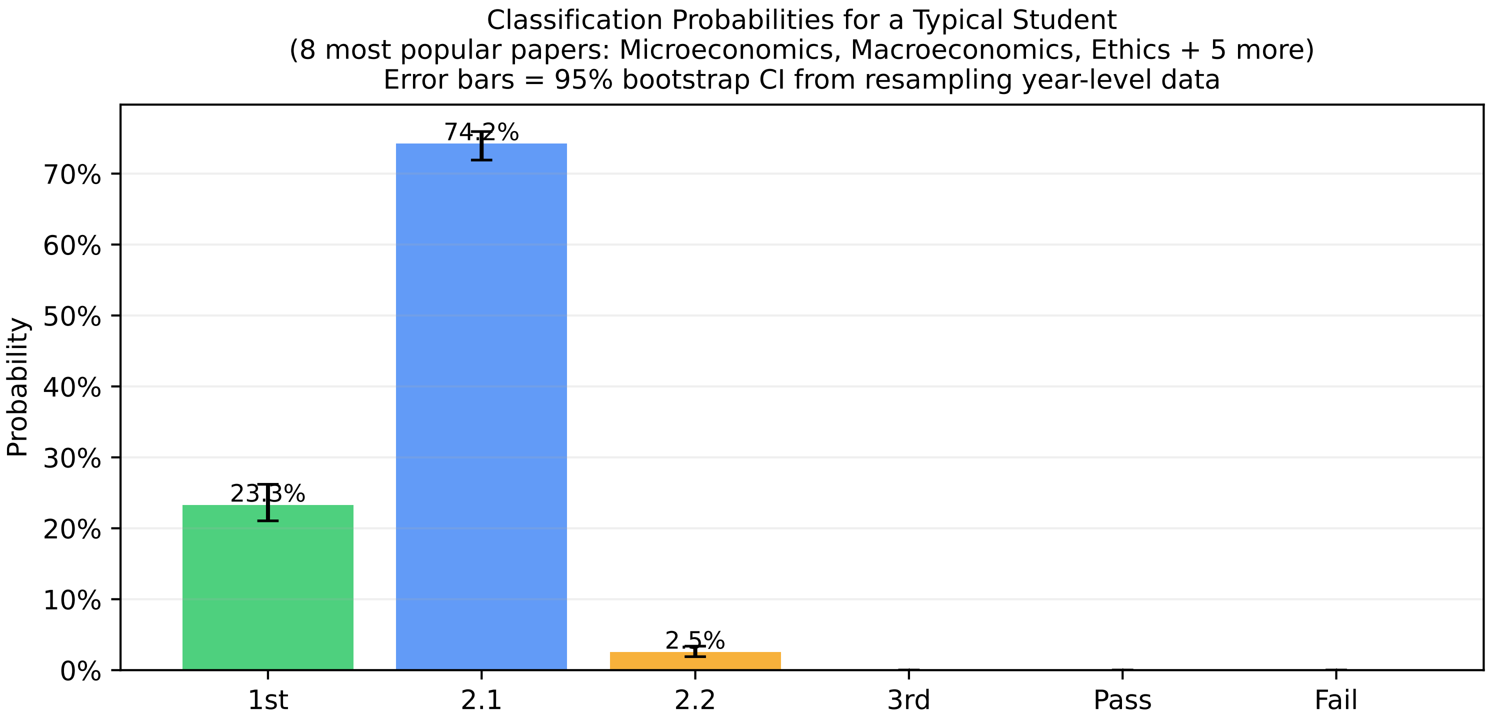

COVID 2020: First Rate Doubled Despite Identical Paper-Level Marks
First-Class Rate (2020 in red)
First rate (%)
Year
0
5
10
15
20
25
30
35
40
2016
2018
2020
2022
2024
Band Distribution: 2020 vs Other Years
% of marks
Other years
2020
0
10
20
30
40
50
60
>=70
60-69
50-59
40-49
30-39
<30



Subject Market Share Over Time
Economics
Philosophy
Politics
Share of paper-sittings (%)
Year
45
40
35
30
25
20
2005.0
2007.5
2010.0
2012.5
2015.0
2017.5
2020.0
2022.5
2025.0



Paper Popularity Trends (share of all paper-sittings, significant trends with >1pp change)
Growing Papers (>1pp total change)
Development of the World Eco (+2.8pp)
Special Subject: Feminist Th (+1.5pp)
British Politics and Governm (+1.2pp)
Share of paper-sittings (%)
Year
Declining Papers (>1pp total change)
International Relations in t (-2.4pp)
Economics of Developing Coun (-1.9pp)
Early Modern Philosophy (-1.8pp)
Comparative Government (-1.2pp)
International Economics (-1.2pp)
International Relations in t (-1.1pp)
Public Economics (-1.1pp)
Share of paper-sittings (%)
Year


Classification Probabilities for a Typical Student
(8 most popular papers: Microeconomics, Macroeconomics, Ethics + 5 more)
Error bars = 95% bootstrap CI from resampling year-level data
Probability
0%
10%
20%
30%
40%
50%
60%
70%
23.3%
74.2%
2.5%
1st
2.1
2.2
3rd
Pass
Fail

Kingmaker Papers (σ ≥ 2× median σ = 11.1; median σ = 5.5)

#	Paper	Subject	μ	σ	Total n
1	Econometrics	Economics	65.5	14.0	170
2	Game Theory	Economics	66.4	12.3	157

Methodology Notes

Distribution fitting

Truncated normal (support [0,100]) fitted via MLE on pooled band counts.

6 bins: ≥70, 60–69, 50–59, 40–49, 30–39, <30.

81 papers fitted (65 band MLE, 16 moment estimates). 2020 excluded.

Goodness of fit: chi-squared test (bins merged where expected < 5).

Classification rules

1st: avg ≥ 68.5, ≥ 2 marks of 70+, no mark < 50.

2.1: avg ≥ 59.0, ≥ 3 marks of 60+.

2.2: avg ≥ 49.0, ≥ 3 marks of 50+.

3rd: avg ≥ 40.0, ≥ 3 marks of 40+.

Monte Carlo simulation

$\text{mark}_i = \mu_i + \theta + \varepsilon_i$, $\theta \sim N(0, \sigma^2_{\text{ability}})$, $\varepsilon_i \sim N(0, \sigma^2_{\text{paper}} - \sigma^2_{\text{ability}})$.

$\sigma_{\text{ability}} = 2.58$, calibrated to 23.4% first-class rate.

100,000 simulations per paper combination.

Temporal trends

OLS regression of mean mark on year, excluding 2020.

95% CI: slope $\pm\, t_{n-2,\ 0.025} \times$ stderr.

Key limitations

- Truncated normal may poorly fit papers with ceiling effects or bimodality.
- Single-factor ability model; same-subject papers may be more correlated.
- Selection effects: observed distributions reflect who chose the paper.
- Temporal pooling: distributions are averages across years, not year-specific.